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FIT HUB

• AYE FIT HUB • DAY 17 OF 30

PROTEIN & MUSCLE

How Much, When, and Why It Matters

Day 17 of 30 | NUT Nutrition | AYE Fit Hub

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- How protein actually builds muscle — the science of MPS explained simply
- How much protein you need: the 1.6–2.2g/kg rule and how to calculate yours
- Protein timing — does when you eat protein actually matter? (research verdict)
- Complete vs incomplete proteins — and why plant eaters need to know the difference
- Best protein sources ranked: eggs, chicken, lentils, dairy, beans, whey
- How to hit your protein target every day without tracking obsessively
- Common protein myths busted — kidneys, too much protein, timing windows

Why Protein Is the Most Important Macronutrient for Athletes

Carbohydrates fuel your workouts. Fats support hormones. But protein is the only macronutrient that can **build, repair, and maintain muscle tissue**. Without adequate protein, your training stimulus — every rep, every sprint, every stretch — produces no lasting adaptation. The muscle damage from exercise requires amino acids (the building blocks of protein) to rebuild stronger. No amino acids = no adaptation = no progress.

Protein is also the most satiating macronutrient — gram for gram, it keeps you fuller longer than carbohydrates or fat. This makes it the single most powerful dietary tool for anyone trying to lose fat while preserving muscle. High-protein diets consistently outperform low-protein diets for body composition changes, regardless of total calorie intake.

How Protein Builds Muscle — The Science Simplified

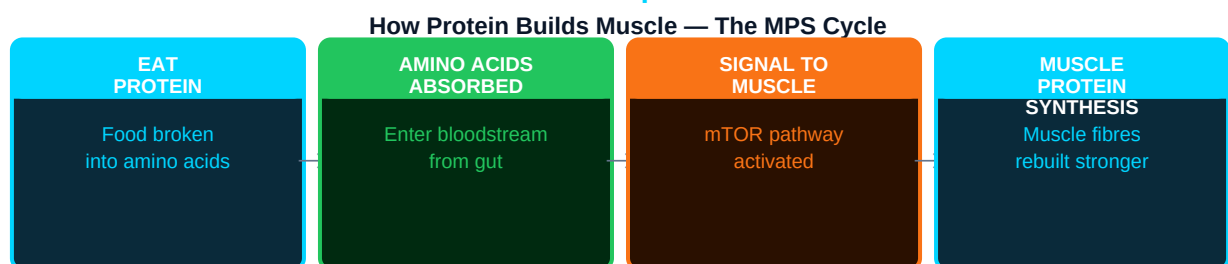


Fig 1 — MPS requires both adequate protein intake AND a training stimulus. Protein alone without exercise does not maximally build muscle.

Muscle Protein Synthesis (MPS) is the biological process by which your body assembles new muscle proteins from amino acids. MPS is stimulated by two things simultaneously: resistance exercise (creates the demand signal) and dietary protein intake (provides the raw materials). Both are required. The key amino acid that triggers MPS is **leucine** — found abundantly in animal proteins, dairy, and whey. A meal needs approximately 2–3g of leucine to maximally stimulate MPS — which is found in about 25–30g of complete protein.

How Much Protein Do You Actually Need?

The most important number in sports nutrition. Research is clear and consistent: the optimal protein intake for people who train regularly is **1.6 to 2.2 grams per kilogram of bodyweight per day (g/kg/day)**. Below 1.6g/kg and muscle gain is suboptimal. Above 2.2g/kg provides no additional muscle-building benefit for most people — though it does no harm either.

Goal	Recommended Intake	Example (75kg person)	Notes
General health (no training)	0.8 g/kg/day (minimum)	60g/day	RDA — adequate for sedentary adults, not optimal for athletes
Recreational training (3×/week)	1.4–1.8 g/kg/day	105–135g/day	Good starting range for most people beginning a training programme
Muscle building (5×/week)	1.6–2.2 g/kg/day	120–165g/day	Research-backed optimal range for maximising MPS and muscle gain

Fat loss (calorie deficit)	2.0–2.4 g/kg/day	150–180g/day	Higher intake during deficit preserves muscle; protein is anti-catabolic
Elite athletes	2.0–3.1 g/kg/day	150–230g/day	Upper range for very high training volumes; marginal extra benefit above 2.2g
Older adults (50+)	1.6–2.4 g/kg/day	120–180g/day	Higher requirement — muscle protein synthesis is less efficient with age (anabolic resistance)

CALCULATE YOURS: $\text{Bodyweight in kg} \times 1.8 = \text{your daily protein target in grams}$. This puts you solidly in the middle of the optimal range for most training goals. Example: $70\text{kg} \times 1.8 = 126\text{g}$ protein per day. Example: $85\text{kg} \times 1.8 = 153\text{g}$ protein per day.

How Much Protein Per Meal?

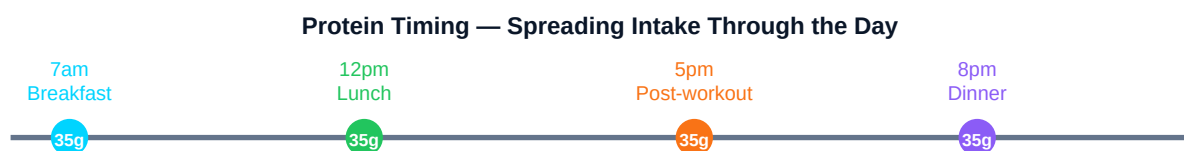


Research shows that **MPS plateaus at approximately 30–40g of protein per meal** for most people. Eating 80g of protein in one meal does not produce twice the muscle-building effect of 40g — the excess amino acids are simply oxidised for energy or converted to other compounds. This is why spreading protein intake across multiple meals is more effective than eating it all at once.

Practical implication: aim for **30–40g of protein in every meal**, distributed across 3–5 meals per day. This maximises the number of MPS-stimulating events in a 24-hour period. For a person needing 150g protein per day: five meals of 30g each, or four meals of 37g each.

When Should You Eat Protein? — Timing Demystified

The "anabolic window" — the idea that you must consume protein within 30 minutes of finishing a workout or you lose all your gains — is one of the most overstated concepts in fitness. The research reality is more nuanced and more practical.



Total: 4 meals × 35g = 140g protein per day | For a 75kg person: 1.87g/kg — within the optimal 1.6–2.2g/kg range

Fig 3 — Spread protein evenly across 3–5 meals rather than eating it all at once for maximum MPS stimulation throughout the day.

Timing Question	The Research Verdict	Practical Implication
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Post-workout protein window — how urgent?	The "window" is approximately 4–6 hours, not 30 minutes. If you ate protein 2–3 hours before training, post-workout urgency is minimal.	Eat protein within 2 hours after training. Immediately is fine if convenient, but not urgently required.
Pre-workout protein — helpful?	Yes — pre-workout protein (2–3 hours before) primes amino acid availability during training, possibly more important than post-workout.	Include protein in your pre-workout meal (2–3 hrs before). Eggs, Greek yoghurt, or chicken are ideal.
Before sleep — worth it?	Casein protein (from dairy, cottage cheese) digests slowly overnight and stimulates MPS during sleep. Meta-analysis shows 40g casein before bed improves overnight muscle recovery.	Eat cottage cheese, Greek yoghurt, or drink milk before sleep. Inexpensive and practical for most people.
Morning protein — important?	After an overnight fast, the body is in net protein breakdown. A protein-rich breakfast stimulates MPS and reverses this catabolic state.	Eat protein at breakfast — eggs, dairy, or protein-rich foods. Do not skip breakfast if muscle gain or maintenance is a goal.
Even distribution vs front-loading	Even distribution (roughly equal protein in each meal) consistently outperforms eating most protein in one or two meals.	Aim for 30–40g protein in every meal rather than having one large protein meal and two small ones.

BOTTOM LINE ON TIMING: Total daily protein intake matters far more than precise timing. If you consistently hit 1.6–2.2g/kg/day spread across 3–5 meals, the exact timing within ± 2 hours has minimal impact on results. Stop obsessing over the clock and focus on hitting your daily total.

Best Protein Sources — Complete and Incomplete

Not all protein sources are equal. The quality of a protein source is determined by its **amino acid profile** (does it contain all 9 essential amino acids?) and its **digestibility** (how efficiently does the body absorb and use it?). Complete proteins contain all 9 essential amino acids in adequate amounts. Incomplete proteins are missing or low in one or more essential amino acids.

Source	Protein per 100g	Complete ?	Leucine per 100g	Digestibility	Practical Notes
Whey protein (powder)	80–90g	YES	~10g	Very high (PDCAAS 1.0)	Fastest-absorbing protein. Ideal post-workout. High leucine triggers MPS maximally.
Eggs (whole)	13g	YES	~1.1g	Very high (PDCAAS 1.0)	Most nutrient-dense protein food. Yolk contains fat-soluble vitamins. 6g protein each.
Chicken breast	31g	YES	~2.4g	High	Lean, versatile, affordable. 150g portion = ~46g protein. Cook in bulk for the week.

Canned tuna/fish	25–30g	YES	~2.2g	High	Cheapest high-quality protein available. 1 can (185g drained) ≈ 40g protein.
Greek yoghurt (full fat)	10g	YES	~0.9g	High	Casein-dominant — slow digesting. Excellent before sleep. Also provides calcium and probiotics.
Cottage cheese	11g	YES	~1.0g	High	High in casein. 200g serving = 22g protein. Cheap, filling, excellent overnight protein source.
Lentils (cooked)	9g	INCOMPL ETE	~0.65g	Medium (lower)	Low in methionine. Combine with rice for complete amino acid profile. High in fibre.
Chickpeas/beans (cooked)	8–9g	INCOMPL ETE	~0.6g	Medium	Combine with whole grains. High fibre content increases satiety significantly.
Tofu (firm)	8g	YES	~0.6g	Medium-high	One of the best plant proteins — complete amino profile. 150g = 12g protein. Low leucine vs animal sources.
Tempeh	19g	YES	~1.4g	High	Fermented soybean — higher protein than tofu, better digestibility. Excellent for plant-based athletes.
Quinoa (cooked)	4g	YES	~0.3g	Medium	One of the only complete plant protein grains. Low total protein — use as complement, not primary source.

FOR PLANT-BASED ATHLETES: You can absolutely build muscle on a plant-based diet, but you must eat more total protein (aim for 2.0–2.4g/kg rather than 1.6g/kg) to compensate for lower digestibility and leucine content of plant proteins. Combine different sources throughout the day to cover all essential amino acids. Consider a leucine supplement (5g) or soy/pea protein powder to boost MPS signalling.

How to Hit Your Protein Target Every Day — Practical Guide

The biggest reason people fail to meet their protein target is not knowledge — it is planning. Protein-rich foods require more preparation than carbohydrate foods. Here is a practical system that works for busy people:

Strategy	How to Apply It	Protein Gained
Anchor every meal with protein FIRST	Before adding carbs or fats to your plate, put the protein source down first. This ensures protein is always present, not forgotten or reduced if portions get adjusted.	Consistent 30–40g per meal

Batch cook protein once per week	Cook 1kg of chicken breast, a dozen boiled eggs, or a large pot of lentils on Sunday. Pre-portioned protein removes all preparation barriers during the week.	All weekly protein ready in 1 hour of cooking
Keep high-protein snacks accessible	Greek yoghurt, cottage cheese, hard-boiled eggs, canned tuna, and edamame require zero preparation. These fill gaps between meals easily.	Extra 20–40g/day without effort
Add protein powder strategically	Use whey or plant protein powder only to fill genuine gaps — not as a replacement for real food. One shake adds 25g protein. Cost: much less than food per gram of protein.	25g per serving when needed
Drink milk or eat dairy daily	Full-fat milk (3.5g protein per 100ml), Greek yoghurt, and cottage cheese are the easiest, cheapest, most nutritious protein sources available. 500ml milk = 17g protein.	15–25g/day with minimal effort
Eat eggs at breakfast	Three whole eggs at breakfast = 18g protein, all essential amino acids, and fat-soluble vitamins. The single fastest high-protein breakfast available. Add two egg whites for 26g total.	18–26g in under 5 minutes cooking time

Common Protein Myths — Busted

Myth	The Truth
"High protein damages your kidneys"	FALSE for healthy people. This myth comes from research on patients who already had kidney disease — protein restriction is appropriate for them. In people with normal kidney function, even very high protein intakes (3+ g/kg) show no evidence of kidney damage in research up to 2 years. If you have diagnosed kidney disease, consult your doctor before increasing protein.
"You can only absorb 30g of protein per meal"	FALSE. The body absorbs all the protein you eat — the absorption limit is approximately your total gut capacity. The 30g "rule" refers to the maximum dose for STIMULATING MPS, not absorbing protein. Excess protein beyond MPS needs is simply oxidised for energy.
"Plant protein is inferior and cannot build muscle"	FALSE — with caveats. Plant proteins are lower in leucine and digestibility, making them less efficient per gram. But consuming more plant protein (2.0–2.4g/kg vs 1.6g/kg) and combining sources closes the gap. Elite plant-based athletes routinely build significant muscle mass.
"Protein makes you fat"	FALSE — protein is the macronutrient least likely to cause fat gain. It has the highest thermic effect of food (25–30% of its calories are burned in digestion) and is the most satiating macronutrient. High protein diets consistently produce better fat loss outcomes than lower-protein diets at the same calories.
"You must eat protein immediately after training or lose your gains"	MOSTLY FALSE. As discussed in Section 4, the anabolic window is 4–6 hours, not 30 minutes. If you ate protein before training, post-workout timing is nearly irrelevant. Total daily intake matters far more than precise timing.

Your Protein Action Plan — Starting Today

Step	Action to Take
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Calculate your target	Bodyweight in kg $\times$ 1.8 = daily grams. Write this number down. This is your daily non-negotiable target.
Plan 4 protein anchors	Choose 4 meals today. Assign a protein source to each: e.g. eggs at breakfast, tuna at lunch, Greek yoghurt as snack, chicken at dinner.
Add up today's total	Count the grams in your 4 planned meals. If you are short, identify where to add: extra eggs, a scoop of protein powder, or a glass of milk.
Batch cook this weekend	Cook 1kg of your main protein source (chicken, lentils, eggs). Portion into 150g servings. All week's protein is now ready in 2 minutes per meal.
Track for 7 days only	Use any app (Cronometer, MyFitnessPal) for 7 days to understand what hitting your target looks and feels like. After 7 days you will not need to track anymore.

Day 18 Preview: Lower Body Strength — Squats, Lunges, and Hip Hinge. Tomorrow we return to training and tackle the most important lower body movement patterns for building strength and athletic power from the ground up.